

CALGARY INTL, AB, Canada

WMO: 718770

Lat: 51.114N

Lon: 114.020W

Elev: 3557

StdP: 12.90

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -17.9	(c) -12.0	(d) -26.5	(e) 1.4	(f) -17.7	(g) -20.3	(h) 2.0	(i) -11.7	(j) 31.7	(k) 33.8	(l) 28.7	(m) 30.6	(n) 6.1	(o) 200	(p) 0.692

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	83.8	60.8	80.3	59.9	77.0	58.8	64.1	77.8	62.1	76.1	60.3	73.5	10.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.1	85.4	70.2	56.7	78.3	67.4	54.9	73.1	66.0	31.2	77.7	29.6	76.6	28.3	73.3	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a) 26.3	(b) 22.8	(c) 19.9	(e) -23.4	(f) 89.7	(g) 6.3	(h) 3.0	(i) -27.9	(j) 91.8	(k) -31.6	(l) 93.6	(m) -35.1	(n) 95.3	(o) -39.6	(p) 97.5
			(e) -23.8	(f) 68.0	(g) 6.2	(h) 3.1	(i) -28.2	(j) 70.3	(k) -31.8	(l) 72.1	(m) -35.3	(n) 73.8	(o) -39.8	(p) 76.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.3	19.6	21.0	28.4	39.7	49.7	56.6	62.7	61.1	52.9	41.3	28.7	20.3	
	DBStd	19.07	16.07	14.18	14.24	9.35	8.32	5.81	5.43	6.01	8.29	9.17	12.79	14.30	
	HDD50	4783	941	811	670	325	108	10	1	3	63	290	639	922	
	HDD65	9098	1406	1231	1134	760	477	258	108	148	367	734	1088	1387	
	CDD50	1241	0	0	1	15	98	210	396	348	151	21	1	0	
	CDD65	78	0	0	0	0	2	7	38	27	4	0	0	0	
	CDH74	1394	0	0	0	4	65	106	557	499	160	3	0	0	
	CDH80	319	0	0	0	0	5	16	140	127	31	0	0	0	
Wind	WSAvg	8.7	8.5	8.2	8.9	9.7	9.6	9.1	8.5	7.9	8.1	8.7	8.3	8.3	
Precipitation	PrecAvg	16.7	0.5	0.4	0.7	1.1	2.3	3.4	2.8	2.0	1.8	0.7	0.6	0.5	
	PrecMax	24.8	1.1	1.3	2.3	3.1	5.7	7.0	6.3	6.6	5.8	1.9	1.7	1.5	
	PrecMin	11.5	0.0	0.0	0.2	0.0	0.4	0.5	0.4	0.4	0.2	0.0	0.0	0.0	
	PrecStd	2.7	0.3	0.3	0.4	0.7	1.1	1.7	1.4	1.3	1.3	0.4	0.4	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.7	54.0	61.9	71.9	80.1	82.3	88.1	88.3	84.3	72.0	61.1	53.9	
		MCWB	40.3	40.3	44.3	50.1	55.0	60.7	63.5	60.9	57.6	51.7	45.3	38.5	
	2%	DB	48.0	48.4	56.0	66.2	75.2	76.9	84.0	83.5	78.7	66.0	54.5	47.5	
		MCWB	37.3	36.9	41.5	47.3	54.0	58.2	62.1	60.3	57.0	49.4	41.9	36.7	
	5%	DB	44.7	43.9	51.6	60.8	70.3	73.1	80.2	79.7	73.8	60.7	49.9	43.6	
		MCWB	35.3	34.4	39.2	44.5	51.7	56.6	62.1	59.7	55.2	46.4	38.9	34.5	
	10%	DB	41.1	40.1	47.1	55.7	66.1	69.8	76.5	75.7	69.0	56.4	45.4	39.7	
		MCWB	33.2	31.9	36.8	41.8	49.9	54.9	60.6	58.8	53.1	43.9	36.3	31.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.8	41.0	45.7	51.0	57.4	63.7	68.3	65.2	60.2	53.5	46.1	40.4
			MCDB	51.0	53.6	59.7	69.0	74.4	77.5	79.7	79.6	78.2	68.6	57.8	50.8
2%		WB	37.8	37.4	42.4	48.1	54.9	60.5	65.4	62.9	57.6	49.9	42.5	37.3	
		MCDB	47.0	47.6	54.9	64.5	71.9	72.5	78.3	77.9	75.0	64.0	53.4	47.1	
5%		WB	35.7	34.7	39.6	45.5	52.8	58.3	63.3	61.1	55.8	47.1	39.7	34.6	
		MCDB	44.3	43.3	50.4	59.4	68.0	70.3	76.9	75.8	72.0	59.4	49.3	43.1	
10%		WB	33.2	32.1	36.9	42.7	50.6	56.3	61.4	59.4	53.8	44.5	36.8	32.0	
		MCDB	41.0	39.5	46.5	54.0	64.2	67.3	74.3	73.4	68.1	55.2	44.9	39.5	
Mean Daily Temperature Range	5% DB	MDBR	18.9	18.9	19.0	20.6	21.3	19.8	22.2	23.0	22.3	20.4	18.0	18.1	
		MCDBR	22.6	24.2	26.8	30.2	29.0	26.0	28.9	31.2	32.2	27.9	24.6	22.2	
		MCWBR	14.9	15.6	15.7	16.0	14.0	12.5	12.8	13.3	15.0	15.0	14.8	14.6	
	5% WB	MCDBR	22.2	23.8	25.8	28.5	26.3	23.5	25.6	26.7	29.8	26.9	24.6	22.4	
		MCWBR	15.0	15.7	15.8	15.3	13.1	12.2	12.9	12.5	14.3	14.9	15.2	15.2	
Clear-Sky Solar Irradiance	taub	0.251	0.268	0.280	0.333	0.341	0.344	0.340	0.356	0.312	0.278	0.253	0.236		
	taud	2.414	2.436	2.477	2.335	2.376	2.406	2.419	2.385	2.475	2.550	2.498	2.409		
	Ebn at Noon	254	280	297	288	290	289	288	277	280	273	254	243		
	Edh at Noon	18	23	28	36	36	36	34	34	28	21	17	16		
All-Sky Solar Radiation	RadAvg	358	669	1074	1478	1749	1868	2052	1678	1183	728	389	279		
	RadStd	25	52	79	107	133	134	124	117	127	77	29	22		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+131	N/A
(47) Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page